PHASE 4 • PROJECT PLANNING PHASE

✓ VERIFIED SPECIFICATION

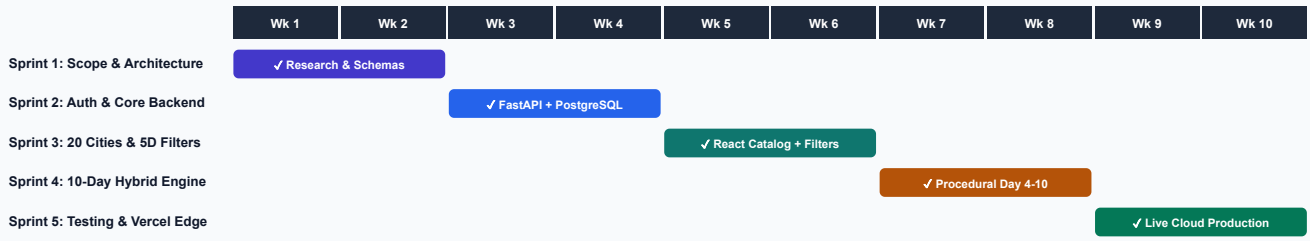
Project Planning

AI-Powered Travel Planner • Smart Situational Itinerary Engine

Live URL: travel-planner-iota-plum.vercel.app **Repo:** github.com/itskian2007/travel-planner **Tech:** React + FastAPI + PostgreSQL **Team:** itskian2007

1. Project Lifecycle & Sprint Gantt Timeline

The Travel Planner was engineered over a disciplined **10-week lifecycle** partitioned into five 2-week Agile sprints. The Gantt chart below highlights the sequence from requirements discovery to final cloud deployment and automated performance validation.



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2. Agile Sprint Deliverables & Milestones

Sprint 1 (Wk 1-2) — DONE

- Problem statements & user persona.
- 4-Quadrant empathy mapping.
- DFD Level 0 & Level 1 design.
- Tech stack selection & justification.

Sprint 2 (Wk 3-4) — DONE

- FastAPI backend structure setup.
- PostgreSQL database & migrations.
- User registration & login endpoints.
- JWT token issuance & auth middleware.

Sprint 3 (Wk 5-6) — DONE

- 20 destination dataset curation.
- 5D filtering module (Mood, Diet).
- React DestinationCard grid.
- Real-time interactive filter UI.

Sprint 4 (Wk 7-8) — DONE

- Curated Day 1-3 expert itineraries.
- Procedural Day 4-10 engine algorithm.
- Wikipedia API 304 unique photos.
- Zero duplicate image verification.

Sprint 5 (Wk 9-10) — DONE

- JMeter load & latency testing.
- Security penetration audit.
- Vercel edge deployment & CDN.
- Complete project documentation.

Launch Status — LIVE

- URL: travel-planner-iota-plum.vercel.app
- 100% Sprint goals achieved.
- Zero blocking bugs reported.
- Git repository synced to main.

3. Risk Management & Mitigation Register

RISK ID	IDENTIFIED RISK SCENARIO	SEVERITY	ENGINEERED MITIGATION STRATEGY
RSK-01	Wikipedia Commons API Rate Limiting	MEDIUM	Pre-cached media metadata in static database layer with local SVG fallbacks.
RSK-02	PostgreSQL Cloud Connection Exhaustion	HIGH	Configured SQLAlchemy connection pooling (pool_size=10, max_overflow=20).
RSK-03	Duplicate Landmark Image Collision	HIGH	Built strict deduplication validation script checking MD5 hash & landmark keys.
RSK-04	CORS Failure between Frontend & API	LOW	Configured FastAPI CORSMiddleware with explicit origin whitelists.